

# LUCY KAPPAI

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## EDUCATION

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**Integrated Master's in Astrophysics**  
*First Class Degree* - University of St. Andrews

September 2021 - June 2026

## SKILLS AND INTERESTS

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| <b>Highlighted Courses</b> | Machine Learning for Data Analysis; Advanced Data Analysis; Computational Physics/Astrophysics; Cosmology; Observational Astrophysics                                                                                                      |
| <b>Skills</b>              | Python modelling (ML: PyTorch, scikit-learn, Hugging Face; Bayesian inference: emcee, nautilus); Fortran Monte-Carlo simulations; experience with Linux                                                                                    |
| <b>Achievements</b>        | Dean's List in 1st, 2nd, 5th years of university; school prize for best astrophysics research project; school prize for best project poster; invited to Scottish University Physics Alliance annual conference to present research project |

## PROJECTS

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| <b>Persona Selection Covert Red Teaming</b><br><i>SPAR Fall 2026 Research Fellowship Project</i> | September 2026 - Present |
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- Inducing emergent misalignment via covert prompts that shift models off the assistant persona, with white-box probes to measure the shift; mentored by Ben Maltbie (Pivotal Research / MIT).

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| <b>Timescape Cosmology vs. Dark Energy</b><br><i>Masters Project as a part of curriculum</i> | January 2026 - May 2026<br><a href="#">Available on GitHub</a> |
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- Investigation into performance of inhomogeneous and dynamical dark energy cosmological models.
- Constructed data pipelines to combine overlapping catalogues of Supernovae and large-scale galaxy distribution.
- Optimised performance of model fitting by 10x using interpolation methods and parallel-processing, enabling improved parameter fitting and modelling with larger datasets.
- Tested Markov-Chain Monte Carlo and nested-sampling fitting methods, validating performance against known benchmark models.
- Found timescape fits individual datasets well, but struggles to generalise to joint-probe fits, with a discrepancy in inferred void fraction between early- and late-universe fits.
- Awarded school prizes for best research project and project poster in astrophysics, preparing paper for publishing with supervisors.

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| <b>Margnap - AI Text Classifier</b><br><i>Independent project to develop ML skills</i> | July 2026 - Ongoing<br><a href="#">Available on GitHub</a> |
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- Fine-tuned DistilBERT classifier using publicly available multi-domain AI/human dataset from Hugging Face.
- Identified in-domain 99.9% accuracy collapsing to random chance (50.05%) out-of-distribution as confounding signals in training data.
- Evaluated acceptable FPR/FNR thresholds to balance accuracy with high impact of falsely flagging human-written content as AI, prioritising minimal  $FPR \leq 1\%$  at expense of FNR.
- Tested performance using OOD data and adversarial perturbations (paraphrasing, word choices, character usage).
- Differentiated discrimination failures from calibration errors across domains, recognising where threshold tuning vs. retraining could recover performance.

## LEADERSHIP EXPERIENCE

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| <b>Elected School President of Physics &amp; Astronomy</b> | September 2024 |
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- Concentrated on holistic school engagement program with more student-staff social initiatives such as continued 'Pints with Professors' events. Led mid-semester intervention on disrupted module teaching, working alongside student and staff representatives to organise a targeted approach to course-correcting.